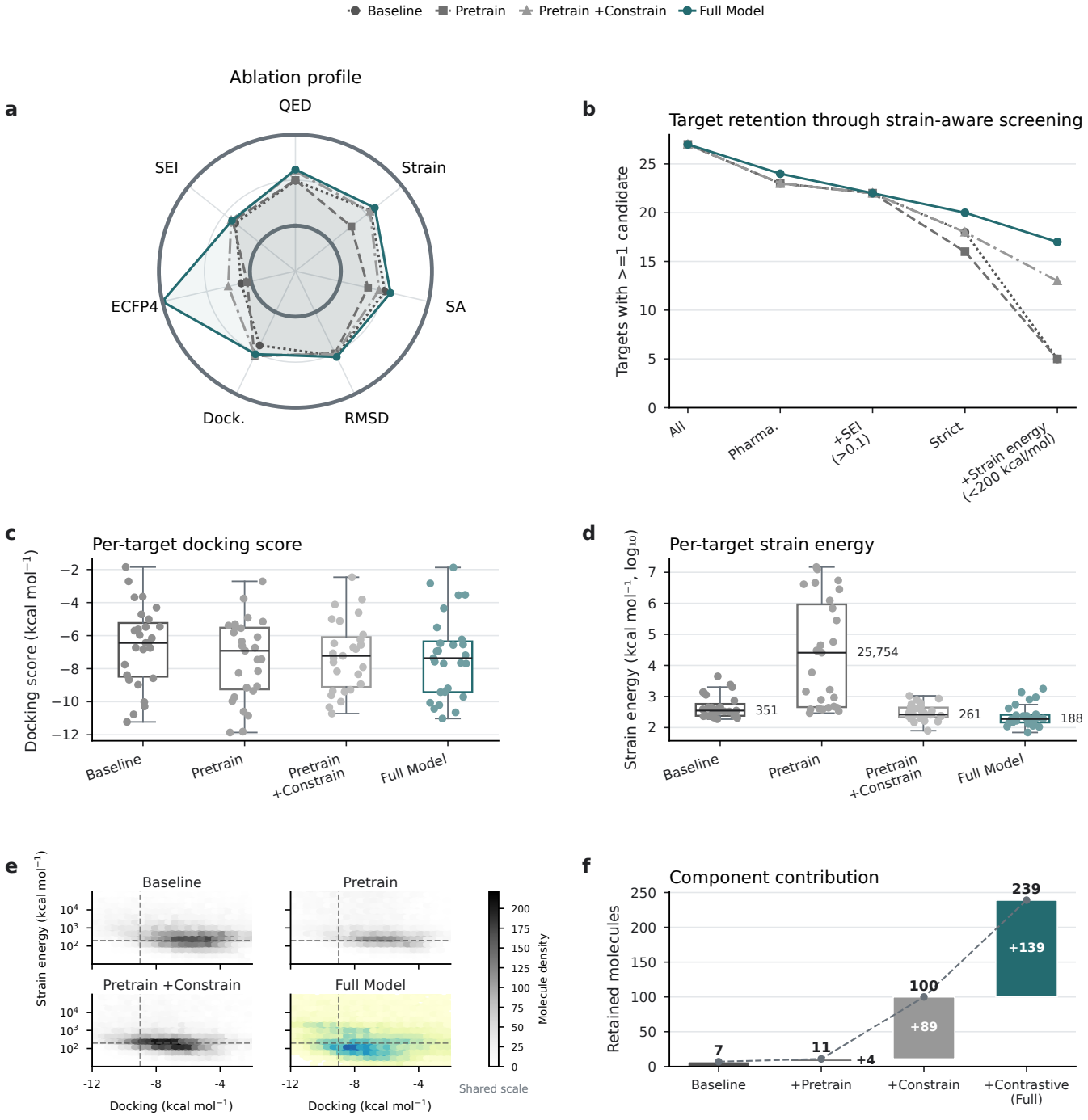

Baseline
Pretrain
Pretrain +Constrain
Full Model
a
Ablation profile
QED
Strain
SA
RMSD
Dock.
ECFP4
SEI
b
Target retention through strain-aware screening
Targets with >=1 candidate
All
Pharma.
+SEI (>0.1)
Strict
+Strain energy (<200 kcal/mol)
c
Per-target docking score
Docking score (kcal mol⁻¹)
Baseline
Pretrain
Pretrain +Constrain
Full Model
d
Per-target strain energy
Strain energy (kcal mol⁻¹, log₁₀)
351
25,754
261
188
e
Baseline
Pretrain
Pretrain +Constrain
Full Model
Strain energy (kcal mol⁻¹)
Docking (kcal mol⁻¹)
Molecule density
Shared scale
f
Component contribution
Retained molecules
7
11
+4
100
+89
239
+139
Baseline
+Pretrain
+Constrain
+Contrastive (Full)